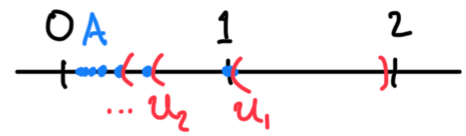
<u>Last time</u>: • $A \subseteq X$ compact $\iff$ every open cover has a finite subcover

• Any finite set is compact

<u>Ex</u> $(\mathbb{R}, |\cdot|)$

① $A = \{\frac{1}{n} : n \in \mathbb{N}\}$ is not compact:

Consider $\mathcal{U}_n = (\frac{1}{n}, 2)$ $\forall n \in \mathbb{N}$.



Then $\{\mathcal{U}_n\}_{n \in \mathbb{N}}$ is an open cover:

$$\frac{1}{n} \in \left(\frac{1}{n+1}, 2\right) \implies \frac{1}{n} \in \bigcup_{m \in \mathbb{N}} \mathcal{U}_m \quad \forall n \in \mathbb{N}$$

Suppose $J \subseteq \mathbb{N}$ is finite. Set $N = \max\{j : j \in J\}$.

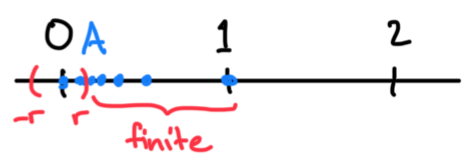
Then

$$\frac{1}{N} \notin \left(\frac{1}{N}, 2\right) \supseteq \mathcal{U}_j \quad \forall j \in J \implies \frac{1}{N} \notin \bigcup_{j \in J} \mathcal{U}_j$$

So there is no finite subcover of A.

② $A \cup \{0\}$ is compact: Suppose $\{\mathcal{U}_i\}_{i \in I}$ is an open cover of $A \cup \{0\}$. Then

$$0 \in \bigcup_{i \in I} \mathcal{U}_i \implies \exists i_0 \text{ s.t. } 0 \in \underset{\substack{\uparrow \\ \text{open}}}{\mathcal{U}_{i_0}}$$



$$\implies \exists r > 0 \text{ s.t. } (-r, r) \subseteq \mathcal{U}_{i_0}$$

Let $N \in \mathbb{N}$ s.t. $\frac{1}{N} < r$. Then $\{\frac{1}{n} : n = 1, 2, \ldots, N-1\}$ is finite, so by the previous example $\exists i_1, \ldots, i_{N-1} \in I$ s.t. $\frac{1}{n} \in \mathcal{U}_{i_n}$ for each $n = 1, 2, \ldots, N-1$. Altogether,

$J = \{i_0, i_1, \ldots, i_{N-1}\}$ is finite and $\{\mathcal{U}_i\}_{i \in J}$ is a finite subcover of $A \cup \{0\}$.

<u>Prop</u> Let (X, d_X) be a metric space, (Y, d_Y) a subspace, and $A \subseteq Y \subseteq X$. Then: A is compact in $(X, d_X) \iff A$ is compact in (Y, d_Y).

<u>Pf</u>: <u>$\Rightarrow$</u>|: Suppose A is compact in X. Let $\{V_i\}_{i \in I}$ be a cover of A with V_i open in Y $\forall i \in I$. By Lecture 24, $\exists\, \mathcal{U}_i$ open in X s.t. $V_i = \mathcal{U}_i \cap Y$. Then $V_i \subseteq \mathcal{U}_i$ $\forall i \in I$, so $\{\mathcal{U}_i\}_{i \in I}$ is a cover of A that is open in X. As A is compact in X, $\exists$ a finite subcover $\{\mathcal{U}_i\}_{i \in J}$. Then $\{V_i\}_{i \in J}$ is a finite subcover of A as well, since

$$A \subseteq \left(\bigcup_{i \in J} \mathcal{U}_i\right) \cap Y = \bigcup_{i \in J} (\mathcal{U}_i \cap Y) = \bigcup_{i \in J} V_i .$$

So A is compact in Y.

<u>$\Leftarrow$</u>|: Suppose A is compact in Y. Let $\{\mathcal{U}_i\}_{i \in I}$ be a cover of A with $\mathcal{U}_i$ open in X $\forall i \in I$. By Lecture 24, $V_i = \mathcal{U}_i \cap Y$ is open in Y $\forall i \in I$. So $\{V_i\}_{i \in I}$ is a cover of A that is open in Y:

$$A \subseteq \left(\bigcup_{i \in I} \mathcal{U}_i\right) \cap Y = \bigcup_{i \in I} (\mathcal{U}_i \cap Y) = \bigcup_{i \in I} V_i$$

As A is compact in Y, $\exists$ a finite subcover $\{V_i\}_{i \in J}$. As $V_i \subseteq \mathcal{U}_i$ $\forall i$, then $\{\mathcal{U}_i\}_{i \in J}$ is also a cover

of A. So A is compact in X. $\square$

- In particular, taking $Y = A$:

<u>Cor</u> A is compact in $X \iff A$ is compact in A.

<u>Thm</u> Let (X, d) be a metric space. If X is compact, then every sequence in X has a subsequence that converges in X.

<u>Pf</u>: Suppose not: X is compact, and there exists a sequence $\{x_n\}_{n \geq 1}$ s.t. every subsequence does not converge.

Step 1: The set of values $\{x_n : n \in \mathbb{N}\}$ is infinite. Otherwise, $\exists$ a constant subsequence $x_{k_n} = x$ $\forall n \geq 1$ which would converge.

Step 2: The set $\{x_n : n \in \mathbb{N}\}$ has no limit points. Suppose $x \in X$ were a limit point. Then $\forall r > 0$, the intersection of $B_r(x)$ and $\{x_n : n \in \mathbb{N}\} \setminus \{x\}$ is nonempty. Construct a subsequence $\{x_{k_n}\}_{n \geq 1}$ as follows:

- Take $r_1 = 1 \implies \exists\, x_{k_1} \in B_{r_1}(x)$
- Take $r_2 = \min\{\frac{1}{2}, d(x, x_1), \ldots, d(x, x_{k_1})\} > 0$

$\implies \exists\, x_{k_2} \in B_{r_2}(x) \implies x_{k_2} \notin \{x_1, \ldots, x_{k_1}\} \implies k_2 > k_1$

Proceeding inductively, we obtain a subsequence $\{x_{k_n}\}_{n \geq 1}$ s.t.

$$d(x_{k_n}, x) < r_n \leq \frac{1}{n} \longrightarrow 0 \quad \text{as} \quad n \to \infty$$

So $\{x_{k_n}\}_{n\geq 1}$ converges to x — a contradiction.
So $\{x_n : n \in \mathbb{N}\}$ has no limit points.

Step 3: We'll construct an open cover of X. For each $x \in X$, we know x is not a limit point for $\{x_m : m \in \mathbb{N}\}$, so $\exists\, r_x > 0$ s.t. $B_{r_x}(x)$ and $\{x_m : m \in \mathbb{N}\} \setminus \{x\}$ do not intersect.

(To be continued!)